

Let us now find the distance between any two points $P(x_1, y_1)$ and $Q(x_2, y_2)$. Draw PR and QS perpendicular to the x -axis. A perpendicular from the point P on QS is drawn to meet it at the point T (see Fig. 7.5).

Then, $OR = x_1$, $OS = x_2$. So, $RS = x_2 - x_1 = PT$.

Also, $SQ = y_2$, $ST = PR = y_1$. So, $QT = y_2 - y_1$.

Now, applying the Pythagoras theorem in ΔPTQ , we get

$$\begin{aligned} PQ^2 &= PT^2 + QT^2 \\ &= (x_2 - x_1)^2 + (y_2 - y_1)^2 \end{aligned}$$

Therefore,
$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Note that since distance is always non-negative, we take only the positive square root. So, the distance between the points $P(x_1, y_1)$ and $Q(x_2, y_2)$ is

$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2},$$

which is called the **distance formula**.

Remarks :

1. In particular, the distance of a point $P(x, y)$ from the origin $O(0, 0)$ is given by

$$OP = \sqrt{x^2 + y^2}.$$

2. We can also write, $PQ = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$. (Why?)

Example 1 : Do the points $(3, 2)$, $(-2, -3)$ and $(2, 3)$ form a triangle? If so, name the type of triangle formed.

Solution : Let us apply the distance formula to find the distances PQ , QR and PR , where $P(3, 2)$, $Q(-2, -3)$ and $R(2, 3)$ are the given points. We have

$$PQ = \sqrt{(3 + 2)^2 + (2 + 3)^2} = \sqrt{5^2 + 5^2} = \sqrt{50} = 7.07 \text{ (approx.)}$$

$$QR = \sqrt{(-2 - 2)^2 + (-3 - 3)^2} = \sqrt{(-4)^2 + (-6)^2} = \sqrt{52} = 7.21 \text{ (approx.)}$$

$$PR = \sqrt{(3 - 2)^2 + (2 - 3)^2} = \sqrt{1^2 + (-1)^2} = \sqrt{2} = 1.41 \text{ (approx.)}$$

Since the sum of any two of these distances is greater than the third distance, therefore, the points P , Q and R form a triangle.

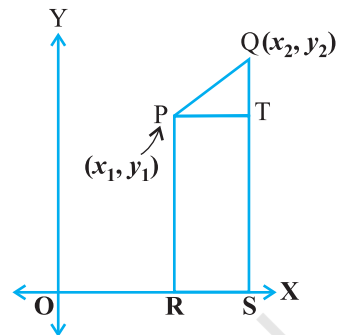


Fig. 7.5

Also, $PQ^2 + PR^2 = QR^2$, by the converse of Pythagoras theorem, we have $\angle P = 90^\circ$. Therefore, PQR is a right triangle.

Example 2 : Show that the points (1, 7), (4, 2), (-1, -1) and (-4, 4) are the vertices of a square.

Solution : Let A(1, 7), B(4, 2), C(-1, -1) and D(-4, 4) be the given points. One way of showing that ABCD is a square is to use the property that all its sides should be equal and both its diagonals should also be equal. Now,

$$AB = \sqrt{(1 - 4)^2 + (7 - 2)^2} = \sqrt{9 + 25} = \sqrt{34}$$

$$BC = \sqrt{(4 + 1)^2 + (2 + 1)^2} = \sqrt{25 + 9} = \sqrt{34}$$

$$CD = \sqrt{(-1 + 4)^2 + (-1 - 4)^2} = \sqrt{9 + 25} = \sqrt{34}$$

$$DA = \sqrt{(1 + 4)^2 + (7 - 4)^2} = \sqrt{25 + 9} = \sqrt{34}$$

$$AC = \sqrt{(1 + 1)^2 + (7 + 1)^2} = \sqrt{4 + 64} = \sqrt{68}$$

$$BD = \sqrt{(4 + 4)^2 + (2 - 4)^2} = \sqrt{64 + 4} = \sqrt{68}$$

Since, $AB = BC = CD = DA$ and $AC = BD$, all the four sides of the quadrilateral ABCD are equal and its diagonals AC and BD are also equal. Therefore, ABCD is a square.

Alternative Solution : We find the four sides and one diagonal, say, AC as above. Here $AD^2 + DC^2 = 34 + 34 = 68 = AC^2$. Therefore, by the converse of Pythagoras theorem, $\angle D = 90^\circ$. A quadrilateral with all four sides equal and one angle 90° is a square. So, ABCD is a square.

Example 3 : Fig. 7.6 shows the arrangement of desks in a classroom. Ashima, Bharti and Camella are seated at A(3, 1), B(6, 4) and C(8, 6) respectively. Do you think they are seated in a line? Give reasons for your answer.

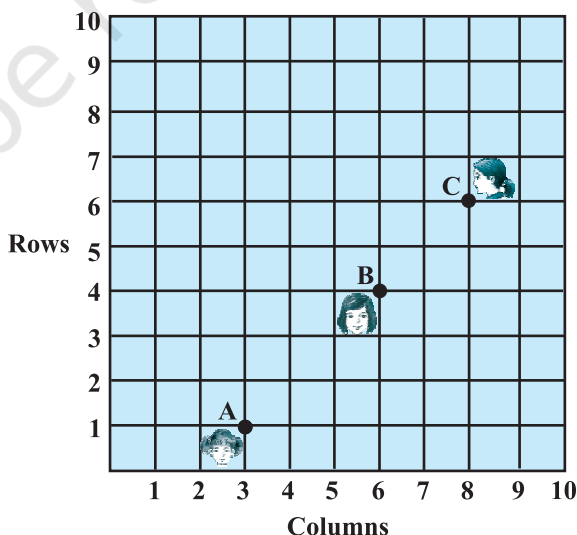


Fig. 7.6

Solution : Using the distance formula, we have

$$AB = \sqrt{(6-3)^2 + (4-1)^2} = \sqrt{9+9} = \sqrt{18} = 3\sqrt{2}$$

$$BC = \sqrt{(8-6)^2 + (6-4)^2} = \sqrt{4+4} = \sqrt{8} = 2\sqrt{2}$$

$$AC = \sqrt{(8-3)^2 + (6-1)^2} = \sqrt{25+25} = \sqrt{50} = 5\sqrt{2}$$

Since, $AB + BC = 3\sqrt{2} + 2\sqrt{2} = 5\sqrt{2} = AC$, we can say that the points A, B and C are collinear. Therefore, they are seated in a line.

Example 4 : Find a relation between x and y such that the point (x, y) is equidistant from the points $(7, 1)$ and $(3, 5)$.

Solution : Let $P(x, y)$ be equidistant from the points $A(7, 1)$ and $B(3, 5)$.

We are given that $AP = BP$. So, $AP^2 = BP^2$

$$\text{i.e.,} \quad (x-7)^2 + (y-1)^2 = (x-3)^2 + (y-5)^2$$

$$\text{i.e.,} \quad x^2 - 14x + 49 + y^2 - 2y + 1 = x^2 - 6x + 9 + y^2 - 10y + 25$$

$$\text{i.e.,} \quad x - y = 2$$

which is the required relation.

Remark : Note that the graph of the equation $x - y = 2$ is a line. From your earlier studies, you know that a point which is equidistant from A and B lies on the perpendicular bisector of AB. Therefore, the graph of $x - y = 2$ is the perpendicular bisector of AB (see Fig. 7.7).

Example 5 : Find a point on the y -axis which is equidistant from the points $A(6, 5)$ and $B(-4, 3)$.

Solution : We know that a point on the y -axis is of the form $(0, y)$. So, let the point $P(0, y)$ be equidistant from A and B. Then

$$(6-0)^2 + (5-y)^2 = (-4-0)^2 + (3-y)^2$$

$$\text{i.e.,} \quad 36 + 25 + y^2 - 10y = 16 + 9 + y^2 - 6y$$

$$\text{i.e.,} \quad 4y = 36$$

$$\text{i.e.,} \quad y = 9$$

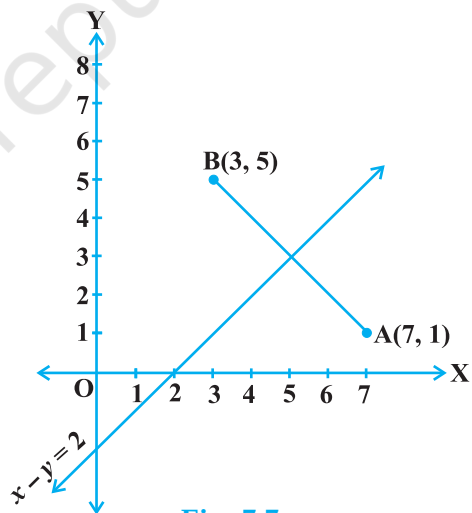


Fig. 7.7

Example 6 : Find the coordinates of the point which divides the line segment joining the points $(4, -3)$ and $(8, 5)$ in the ratio $3 : 1$ internally.

Solution : Let $P(x, y)$ be the required point. Using the section formula, we get

$$x = \frac{3(8) + 1(4)}{3 + 1} = 7, \quad y = \frac{3(5) + 1(-3)}{3 + 1} = 3$$

Therefore, $(7, 3)$ is the required point.

Example 7 : In what ratio does the point $(-4, 6)$ divide the line segment joining the points $A(-6, 10)$ and $B(3, -8)$?

Solution : Let $(-4, 6)$ divide AB internally in the ratio $m_1 : m_2$. Using the section formula, we get

$$(-4, 6) = \left(\frac{3m_1 - 6m_2}{m_1 + m_2}, \frac{-8m_1 + 10m_2}{m_1 + m_2} \right) \quad (1)$$

Recall that if $(x, y) = (a, b)$ then $x = a$ and $y = b$.

So,
$$-4 = \frac{3m_1 - 6m_2}{m_1 + m_2} \quad \text{and} \quad 6 = \frac{-8m_1 + 10m_2}{m_1 + m_2}$$

Now,
$$-4 = \frac{3m_1 - 6m_2}{m_1 + m_2} \quad \text{gives us}$$

$$-4m_1 - 4m_2 = 3m_1 - 6m_2$$

i.e.,
$$7m_1 = 2m_2$$

i.e.,
$$m_1 : m_2 = 2 : 7$$

You should verify that the ratio satisfies the y -coordinate also.

Now,
$$\frac{-8m_1 + 10m_2}{m_1 + m_2} = \frac{-8 \frac{m_1}{m_2} + 10}{\frac{m_1}{m_2} + 1} \quad (\text{Dividing throughout by } m_2)$$

$$= \frac{-8 \times \frac{2}{7} + 10}{\frac{2}{7} + 1} = 6$$

Therefore, the point $(-4, 6)$ divides the line segment joining the points $A(-6, 10)$ and $B(3, -8)$ in the ratio $2 : 7$.

Alternatively : The ratio $m_1 : m_2$ can also be written as $\frac{m_1}{m_2} : 1$, or $k : 1$. Let $(-4, 6)$

divide AB internally in the ratio $k : 1$. Using the section formula, we get

$$(-4, 6) = \left(\frac{3k - 6}{k + 1}, \frac{-8k + 10}{k + 1} \right) \quad (2)$$

So,
$$-4 = \frac{3k - 6}{k + 1}$$

i.e.,
$$-4k - 4 = 3k - 6$$

i.e.,
$$7k = 2$$

i.e.,
$$k : 1 = 2 : 7$$

You can check for the y-coordinate also.

So, the point $(-4, 6)$ divides the line segment joining the points $A(-6, 10)$ and $B(3, -8)$ in the ratio $2 : 7$.

Note : You can also find this ratio by calculating the distances PA and PB and taking their ratios provided you know that A, P and B are collinear.

Example 8 : Find the coordinates of the points of trisection (i.e., points dividing in three equal parts) of the line segment joining the points $A(2, -2)$ and $B(-7, 4)$.

Solution : Let P and Q be the points of trisection of AB i.e., $AP = PQ = QB$ (see Fig. 7.11).

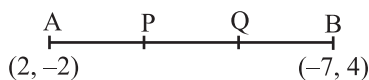


Fig. 7.11

Therefore, P divides AB internally in the ratio $1 : 2$. Therefore, the coordinates of P, by applying the section formula, are

$$\left(\frac{1(-7) + 2(2)}{1 + 2}, \frac{1(4) + 2(-2)}{1 + 2} \right), \text{ i.e., } (-1, 0)$$

Now, Q also divides AB internally in the ratio $2 : 1$. So, the coordinates of Q are

$$\left(\frac{2(-7) + 1(2)}{2 + 1}, \frac{2(4) + 1(-2)}{2 + 1} \right), \text{ i.e., } (-4, 2)$$

Therefore, the coordinates of the points of trisection of the line segment joining A and B are $(-1, 0)$ and $(-4, 2)$.

Note : We could also have obtained Q by noting that it is the mid-point of PB. So, we could have obtained its coordinates using the mid-point formula.

Example 9 : Find the ratio in which the y-axis divides the line segment joining the points $(5, -6)$ and $(-1, -4)$. Also find the point of intersection.

Solution : Let the ratio be $k : 1$. Then by the section formula, the coordinates of the point which divides AB in the ratio $k : 1$ are $\left(\frac{-k + 5}{k + 1}, \frac{-4k - 6}{k + 1} \right)$.

This point lies on the y-axis, and we know that on the y-axis the abscissa is 0.

Therefore,
$$\frac{-k + 5}{k + 1} = 0$$

So,
$$k = 5$$

That is, the ratio is $5 : 1$. Putting the value of $k = 5$, we get the point of intersection as $\left(0, \frac{-13}{3} \right)$.

Example 10 : If the points A(6, 1), B(8, 2), C(9, 4) and D(p, 3) are the vertices of a parallelogram, taken in order, find the value of p.

Solution : We know that diagonals of a parallelogram bisect each other.

So, the coordinates of the mid-point of AC = coordinates of the mid-point of BD

i.e.,
$$\left(\frac{6 + 9}{2}, \frac{1 + 4}{2} \right) = \left(\frac{8 + p}{2}, \frac{2 + 3}{2} \right)$$

i.e.,
$$\left(\frac{15}{2}, \frac{5}{2} \right) = \left(\frac{8 + p}{2}, \frac{5}{2} \right)$$

so,
$$\frac{15}{2} = \frac{8 + p}{2}$$

i.e.,
$$p = 7$$